UI/UX Design Document

GestureX — Bridging Silence, Building Connections

A real-time AI-powered sign language translation application that converts hand gestures into text and speech.

Document Type: UI/UX Design Document
Project: GestureX Mobile Application
Version: 1.0

8. UI/UX Design Document

The UI/UX Design Document describes the proposed user interface and user experience of the GestureX application. It communicates how users will interact with the system, and presents the interface design through annotated screen captures, design rationale, and interaction documentation for each page of the application.

GestureX is a mobile application that uses real-time AI gesture recognition to translate sign language into text and speech, enabling smoother communication between sign-language users and non-signers. The design prioritizes clarity, accessibility, and a minimal number of steps between opening the app and starting a translation session.

8.1 Design Goals

- Provide an accessible, high-contrast interface suitable for a diverse user base, including Deaf and hard-of-hearing users.
- Minimize onboarding friction — from launch to first gesture recognition in as few taps as possible.
- Maintain a consistent visual language (purple brand color, rounded input fields, pill-shaped buttons) across every screen.
- Clearly communicate system status (e.g. camera readiness, recognition state) at all times.
- Reassure users on data privacy at the point where camera access is introduced.

8.2 Visual Design System

Element	Specification
Primary color	Deep purple (#6C2BD9) — buttons, links, active states, logo accent
Secondary color	White backgrounds with soft purple tints for supporting UI
Typography	Bold sans-serif for headings; regular sans-serif for body and form labels
Buttons	Full-width, pill-shaped (fully rounded), filled purple for primary actions
Input fields	Rounded rectangular fields with leading icons (envelope, lock, person, calendar)
Iconography	Simple line icons paired with each input field and status indicator
Imagery	Friendly flat-style illustration used on the welcome screen to build trust

8.3 Application Screen Flow

The core user journey moves linearly from onboarding to the primary gesture recognition feature:

Welcome / Onboarding Screen → Login Screen ■ Create Account Screen → Gesture Recognition Screen (home)

New users land on the Welcome screen, proceed to Create Account (or Login if returning), and are then taken to the Gesture Recognition screen, which functions as the application's home/core screen.

8.4 Screen-by-Screen Design Details

This section documents every screen currently designed in the GestureX Figma prototype, in the order a first-time user experiences them.

8.4.1 Welcome / Onboarding Screen

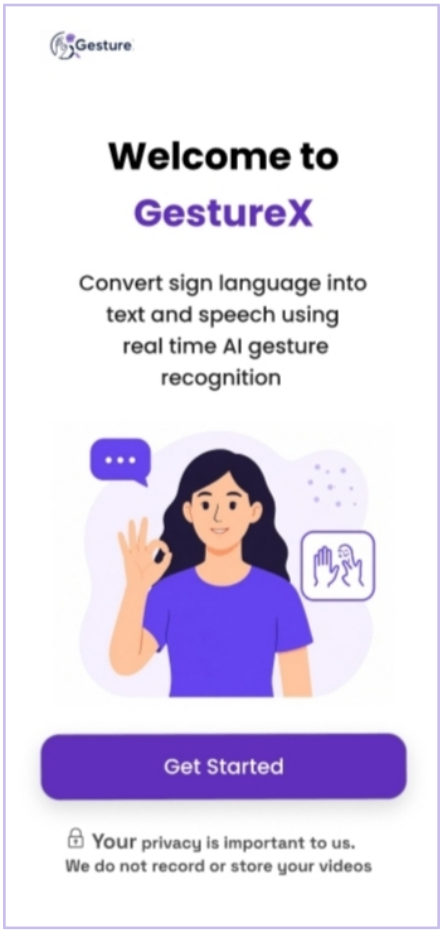


Fig. 8.4.1 — Welcome / Onboarding Screen screen (mobile, portrait)

Purpose

The first screen a new user sees on launching the app. It introduces the GestureX brand and its core value proposition, and sets expectations before requesting camera permissions later in the flow.

Key UI Elements

- Gesture logo (top left)
- Headline: “Welcome to GestureX”
- Sub-headline describing the app's function: converting sign language into text and speech using real-time AI gesture recognition
- Flat-style illustration of a person signing, with a speech-bubble and hand-gesture icon
- Primary button: “Get Started” (full-width, purple, pill-shaped)
- Privacy reassurance line with lock icon: “Your privacy is important to us. We do not record or store your videos.”

Interactions

- Tapping “Get Started” navigates the user to the Login screen.
- The privacy statement is static text (no tap action) intended purely to build trust prior to camera use.

User Flow

This is the entry point of the app and has no back navigation. It is shown once per fresh install / logged-out session, then routes into the Login screen.

Design Notes

The illustration and friendly copy are used to make an AI/camera-based product feel approachable rather than technical, addressing a key trust barrier for first-time users.

8.4.2 Login Screen

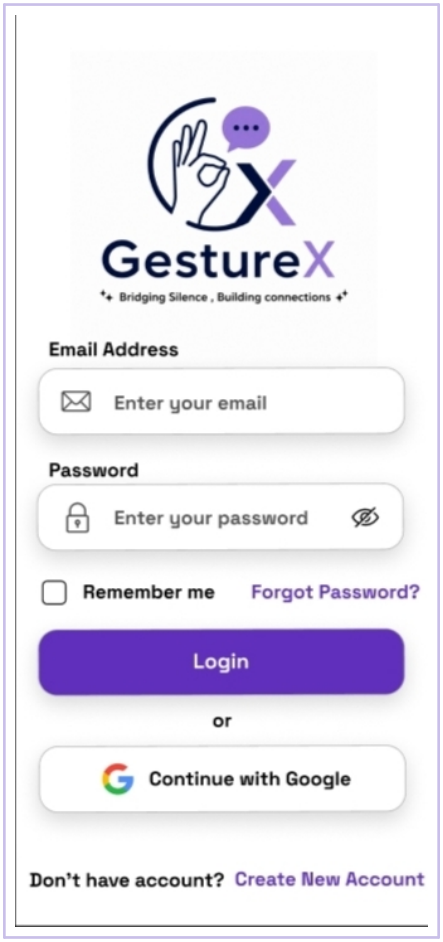


Fig. 8.4.2 — Login Screen screen (mobile, portrait)

Purpose

Allows a returning user to authenticate and access their account. Also acts as the branching point toward account creation for new users.

Key UI Elements

- GestureX logo and tagline: “Bridging Silence, Building Connections”
- Email Address input field with envelope icon and placeholder text
- Password input field with lock icon and a show/hide (eye) toggle
- “Remember me” checkbox
- “Forgot Password?” link (top-right of the password row)
- Primary button: “Login” (full-width, purple, pill-shaped)
- Secondary button: “Continue with Google” (white with Google “G” icon)
- Footer link: “Don't have account? Create New Account”

Interactions

- Entering credentials and tapping “Login” authenticates the user and routes to the Gesture Recognition (home) screen.
- Tapping the eye icon toggles password visibility.
- Tapping “Forgot Password?” opens the password-recovery flow.
- Tapping “Continue with Google” triggers OAuth sign-in as an alternative to email/password.
- Tapping “Create New Account” navigates to the Create Account screen.

User Flow

Reached from the Welcome screen. On successful authentication, the user is taken directly to the core Gesture Recognition screen. Failed login attempts are expected to show inline validation/error messaging beneath the relevant field.

Design Notes

Google Sign-In is offered as a low-friction alternative to reduce sign-up drop-off. The “Remember me” option minimizes repeat logins for frequent users.

8.4.3 Create Account Screen

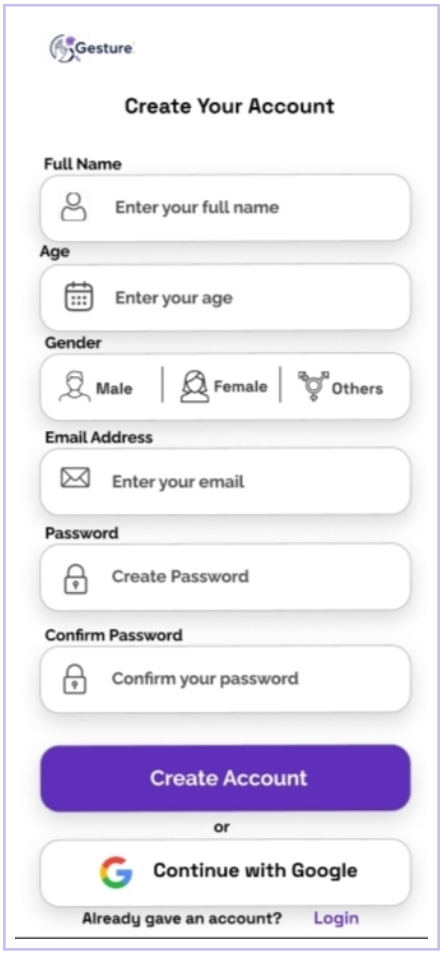


Fig. 8.4.3 — Create Account Screen screen (mobile, portrait)

Purpose

Enables a new user to register for GestureX by providing basic profile information and setting account credentials.

Key UI Elements

- Compact logo at top
- Heading: “Create Your Account”
- Full Name input field with person icon
- Age input field with calendar icon
- Gender selector with three icon-based options: Male, Female, Others
- Email Address input field with envelope icon
- Password field with lock icon (“Create Password”)
- Confirm Password field with lock icon
- Primary button: “Create Account” (full-width, purple, pill-shaped)
- Secondary button: “Continue with Google”
- Footer link: “Already gave an account? Login”

Interactions

- Selecting a Gender option highlights that choice (single-select).
- Tapping “Create Account” validates all fields (including password match) and, on success, creates the account and routes to the Gesture Recognition screen.
- Tapping “Continue with Google” allows account creation via OAuth, skipping manual field entry.
- Tapping “Login” in the footer returns the user to the Login screen.

User Flow

Reached from the Login screen's “Create New Account” link. This is a linear, single-screen form; on successful submission the user proceeds straight into the app's core feature rather than back to Login.

Design Notes

All fields are stacked vertically with consistent field height and iconography, keeping a long form scannable. Password and Confirm Password are visually adjacent to support quick error comparison.

8.4.4 Gesture Recognition Screen (Home)

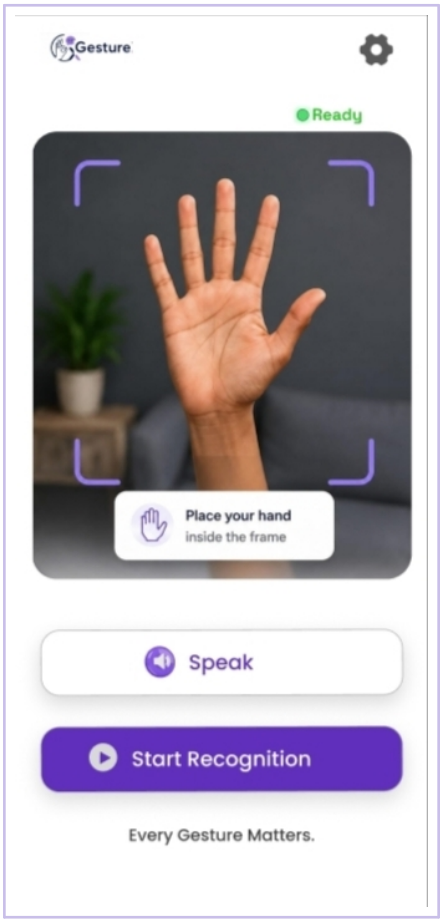


Fig. 8.4.4 — Gesture Recognition Screen (Home) screen (mobile, portrait)

Purpose

The core screen of the application. It activates the device camera and uses real-time AI gesture recognition to detect and translate the user's hand signs into text and speech.

Key UI Elements

- Compact logo (top left) and settings/gear icon (top right)
- System status indicator: “Ready” with a green status dot
- Live camera viewfinder with corner bracket framing guides
- In-frame guidance card: hand icon with instruction “Place your hand inside the frame”
- Secondary button: “Speak” with a speaker/audio icon
- Primary button: “Start Recognition” with a play icon (full-width, purple, pill-shaped)
- Footer tagline: “Every Gesture Matters.”

Interactions

- Tapping the gear icon opens app Settings.
- The status indicator updates (e.g. Ready → Detecting → Recognized) to reflect the current recognition state in real time.
- The guidance card appears when no hand is detected in frame, and is expected to disappear once a hand is correctly positioned.
- Tapping “Start Recognition” begins continuous AI gesture detection and streams recognized signs to on-screen text.
- Tapping “Speak” converts the currently recognized text output into speech using text-to-speech.

User Flow

This is the destination screen after Login or Create Account, and functions as the app's home base. Users are expected to return here for every translation session; the gear icon provides a secondary path out to Settings.

Design Notes

Corner-bracket framing and a live status dot are standard, low-cognitive-load patterns borrowed from camera/scanning UIs, helping users quickly understand how to position their hand without additional instructions.

8.5 Summary

Across all four screens, GestureX maintains a consistent purple-and-white visual identity, full-width pill-shaped primary buttons, and icon-led input fields. The flow is intentionally short — Welcome, then Login/Create Account, then straight into the camera-based Gesture Recognition home screen — minimizing the number of steps between installing the app and translating a first gesture. As the product grows beyond this prototype, this document should be extended with additional pages (e.g. Settings, History/Saved Translations, and Profile) using the same screen-by-screen documentation format established above.